



Shaheed Zulfikar Ali Bhutto Institute of Science & Technology University

DEPARTMENT OF ROBOTICS & ARTIFICIAL INTELLIGENCE

Total Marks: 03

Obtained Marks:

Multivariate Calculus

Assignment # 01

Last Date of Submission: September 28, 2026

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Instructions: Copied or shown assignments will be marked zero. Late submissions are not entertained in any case.

- Show the necessary mathematical steps. Answers without justification may not receive full marks.
- For continuity questions, follow the sequence: check the limit → check the function value → compare.
- If a limit does not exist, clearly state why and then state the conclusion about continuity.

CLO-1

Group 7

- (a) Evaluate the limit by first factoring the numerator:

$$\lim_{(x,y) \rightarrow (2,1)} \frac{(x^2 - 4y^2)}{(x - 2y)}$$

- (b) Find k so that the following function is continuous at (2,1):

$$\begin{cases} \frac{(x^2 - 4y^2)}{(x - 2y)} & \text{if } (x, y) \neq (2, 1) \\ k & \text{if } (x, y) = (2, 1) \end{cases}$$

- (c) Write the continuity condition that determines k.

A)

Given

$$\lim_{(x,y) \rightarrow (2,1)} \frac{(x^2 - 4y^2)}{(x - 2y)}$$

$$x^2 - 4y^2 = (x - 2y)(x + 2y)$$

Factoring the numerator:

$$= (x - 2y)(x + 2y) / (x - 2y)$$

$$= x + 2y$$

Now substitute x=2, y=1

$$= 2 + 2(1) = 4$$

$$= 4 \text{ answer}$$

The limit exists and equals 4.



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B)

$$k = \lim_{(x,y) \rightarrow (2,1)} (x^2 - 4y^2) / (x - 2y)$$

From part (a) the limit is 4.

Therefore:

$$k = 4$$

C)

$$\lim_{(x,y) \rightarrow (2,1)} f(x,y) = f(2,1) \text{ so } k = 4$$